# Chapter 5: Photosynthesis - How Plants Make Food

## 5.1 Introduction

Every living thing needs energy. Animals get their energy by eating plants or other animals, but plants make their own food. The process by which green plants prepare food using sunlight, carbon dioxide and water is called photosynthesis. Because plants produce their own food, they are called autotrophs. Animals and fungi cannot do this and depend on other organisms for food, so they are called heterotrophs.

Photosynthesis takes place mainly in the leaves of a plant. The food produced is glucose, a simple sugar. Plants use some of the glucose immediately for energy and convert the rest into starch, which is stored in leaves, roots, stems and seeds. Photosynthesis also releases oxygen, the gas that most living things need for respiration.

## 5.2 The Overall Equation

The process can be summarised in one chemical equation. Six molecules of carbon dioxide and six molecules of water, in the presence of light and chlorophyll, produce one molecule of glucose and six molecules of oxygen. In symbols this is written as 6CO2 + 6H2O + light energy → C6H12O6 + 6O2. Light energy is converted into chemical energy that is stored inside the glucose molecule.

## 5.3 Raw Materials and Where They Come From

A plant needs three things for photosynthesis: sunlight, carbon dioxide and water. Carbon dioxide enters the leaf from the air through tiny pores called stomata. Each stoma is surrounded by two bean-shaped guard cells that open and close the pore. When the guard cells swell with water the stoma opens, and when they lose water it closes. Oxygen produced during photosynthesis leaves the leaf through the same pores.

Water is absorbed from the soil by the roots and travels upward to the leaves through tubes called xylem vessels. Sunlight is captured by a green pigment called chlorophyll. Chlorophyll absorbs mostly red and blue light and reflects green light, which is why leaves look green.

## 5.4 The Chloroplast

Photosynthesis happens inside small structures called chloroplasts, which are found in large numbers in the cells of the leaf, especially the palisade layer just below the upper surface. A chloroplast is surrounded by a double membrane. Inside it are stacks of flattened sacs called thylakoids, and a stack of thylakoids is known as a granum. Chlorophyll is embedded in the thylakoid membranes. The fluid that fills the space around the grana is called the stroma.

## 5.5 The Two Stages of Photosynthesis

Photosynthesis is not a single step. It happens in two linked stages, the light-dependent reactions and the Calvin cycle.

The light-dependent reactions take place in the thylakoid membranes. Chlorophyll absorbs light energy, and this energy is used to split water molecules into hydrogen and oxygen. This splitting of water is called photolysis. The oxygen is released as a waste product. The energy captured is stored in two energy-carrying molecules, ATP and NADPH.

The Calvin cycle takes place in the stroma and does not need light directly, although it depends on the ATP and NADPH made in the first stage. In this cycle, carbon dioxide is attached to a five-carbon sugar with the help of an enzyme called RuBisCO. Through a series of steps, the energy from ATP and NADPH is used to turn carbon dioxide into a three-carbon sugar, and these molecules are then combined to form glucose. The Calvin cycle is sometimes called the dark reaction, but it actually runs in daylight because it needs the products of the light reactions.

## 5.6 Factors That Affect the Rate of Photosynthesis

The rate of photosynthesis is not constant. It depends on several factors. As light intensity increases, the rate increases until another factor becomes short. Carbon dioxide concentration works in the same way: more carbon dioxide raises the rate up to a point. Temperature matters because photosynthesis depends on enzymes. The rate rises with temperature up to an optimum, which is usually around 25 to 35 degrees Celsius for many plants, and then drops sharply because high temperatures damage the enzymes. Water is also necessary, and a shortage makes the stomata close, which cuts off the supply of carbon dioxide.

When the rate is held back by whichever factor is in shortest supply, that factor is called the limiting factor. On a bright day with little carbon dioxide, for example, carbon dioxide is the limiting factor, and adding more light will not increase the rate.

## 5.7 Experiments on Photosynthesis

Scientists have shown that light and chlorophyll are needed through simple experiments. To show that light is necessary, a potted plant is kept in the dark for two or three days so that its leaves use up their stored starch. This step is called destarching. One leaf is then partly covered with black paper and the plant is placed in sunlight for several hours. The leaf is boiled in water, soaked in warm alcohol to remove the chlorophyll, and then tested with iodine solution. The part that was exposed to light turns blue-black, which shows that starch is present. The covered part stays brown, showing that no starch was made without light.

In 1771 the English scientist Joseph Priestley showed that plants restore the quality of air. He found that a candle burning in a closed jar went out, but a candle burned normally when a mint plant had been kept in the jar for some days. Later, Jan Ingenhousz showed that this happens only in sunlight and only in the green parts of the plant.

## 5.8 Importance of Photosynthesis

Photosynthesis is the foundation of almost every food chain on Earth. Green plants are producers, and the energy they capture passes to herbivores and then to carnivores. The process also supplies the oxygen in the atmosphere that animals and plants use for respiration. At the same time it removes carbon dioxide from the air, which helps to keep the amount of this greenhouse gas in balance. Fossil fuels such as coal and petroleum are themselves the remains of plants and other organisms that once captured sunlight through photosynthesis millions of years ago.

Protecting forests, grasslands and oceans with photosynthesising algae is therefore important for food supply, for clean air and for the climate.

## Summary

Photosynthesis is the process by which green plants use sunlight, carbon dioxide and water to make glucose and release oxygen. It takes place in chloroplasts and has two stages: the light-dependent reactions in the thylakoids and the Calvin cycle in the stroma. Its rate depends on light intensity, carbon dioxide concentration, temperature and water supply, and the factor in shortest supply limits the rate. Photosynthesis supports food chains, supplies oxygen and removes carbon dioxide from the atmosphere.